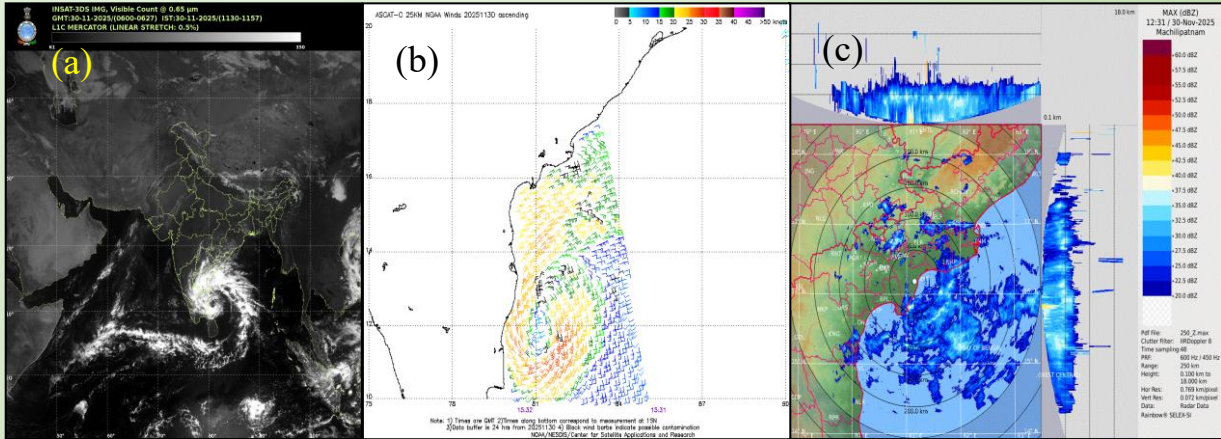




Typical (a) INSAT 3DS visible, (b) ASCAT and (c) Machilipatnam Radar imageries utilised for monitoring cyclonic storm “DITWAH” over Bay of Bengal

Cyclonic Storm “DITWAH” over the Bay of Bengal (26th November– 3rd December, 2025): A Report

Cyclone Warning Division
India Meteorological Department
New Delhi

December, 2025

Cyclonic Storm “DITWAH” over the Bay of Bengal (26th November– 3rd December, 2025): A Report

1. Life History of “DITWAH”:

- An upper air cyclonic circulation lay over Comorin area and neighbourhood in the forenoon (0830 hours IST/ 0300 UTC) of 23rd November, 2025.
- Under its influence, a low pressure area formed over Comorin and adjoining Southwest Bay of Bengal (BoB) & Sri Lanka in the early morning (0530 hrs IST/ 0000 UTC) of 25th November, 2025.
- It lay as well marked low pressure area over southwest Bay of Bengal and adjoining areas of Southeast Sri Lanka & Equatorial Indian Ocean in the early morning (0530 hrs IST/0000 UTC) of 26th November, 2025.
- It concentrated into a depression over southwest BoB and adjoining Sri Lanka in the midnight (2330 hrs. IST/ 1800 UTC) of 26th November, 2025.
- It moved north-northwestwards and intensified into a deep depression over the same region in the early morning (0530 hours IST/ 0000 UTC) and into the cyclonic storm “DITWAH” over the same region in the noon (1130 hours IST/ 0600 UTC) of 27th November.
- Continuing to move to north-northwestwards, it crossed Sri Lanka coast around noon (between 1130 to 1230 hrs. IST/0600 UTC to 0700 UTC) of 27th November as a Cyclonic Storm with the wind speed of 60 to 70 gusting to 80 kmph.
- Thereafter, it continued to move slowly north-northwestwards across coastal Sri Lanka & southwest BoB maintaining the intensity of 60 to 70 gusting to 80 kmph till night of 27th November (2030 hrs. IST/ 1500 UTC) and increase marginally becoming 65-75 gusting to 85 kmph during 27th midnight (2330 hrs. IST/ 1800UTC) to early morning of 29th November (0530 hrs. IST/0000 UTC).
- Thereafter, it emerged into southwest BoB and adjoining North Sri Lanka coasts in the early morning (0530 hrs. IST/0000 UTC) of 29th November. Thereafter, it moved nearly northwards and intensified slightly by reaching the maximum sustained wind speed of 70-80 gusting to 90 kmph during 0830 hrs. IST of 29th to 0530 hrs. IST of 30th November and lay centered about 50 km east of Chennai.
- Continuing to moving further northwards over southwest BoB and adjoining Tamil Nadu-Puducherry coast, it lay about 70km east of north Tamil Nadu-Puducherry coast at 0830 hrs. IST of 30th November and about 30km east of Chennai coast at 1130 hrs. IST of 01st December. While moving northwards, it weakened into a deep depression in the evening (1730 hours IST/ 1200 UTC) of 30th November.
- Thereafter, it remains practically stationary till midnight (2330 hrs. IST/1800 UTC) of 01st December.

- Then it gradually recurved south-southwestwards & weakened into a depression in the early morning (0530 hrs. IST/ 0000 UTC) of 2nd December with its centre about 25 km away from North Tamil Nadu coast and into a well marked low pressure area over North Tamil Nadu-Puducherry coasts and neighbourhood in the early morning (0530 hrs IST/0000 UTC) of 3rd December. The observed track of the system is presented in **Fig.1**. The best track parameters are presented in **Table 1**.

Table 1: Best track positions and other parameters of the Cyclonic Storm “Ditwah” over Southwest Bay of Bengal and adjoining areas of southeast Sri Lanka & Equatorial Indian Ocean during 26th November – 03rd December, 2025

Date	Time (UTC)	Lat. (0N)	Long. (0E)	C.I. No.	ECP (hPa)	ΔP (hPa)	MSW (kt)	Category
26.11.25	1800	5.9	82.6	1.5	1004	4	25	D
27.11.25	0000	6.3	82.4	2.0	1003	5	30	DD
	0300	6.7	82.1	2.0	1003	5	30	DD
	0600	6.9	81.9	2.5	1002	6	35	CS
	0900	7.2	81.8	2.5	1002	6	35	CS
	1200	7.6	81.6	2.5	1002	6	35	CS
	1500	7.7	81.5	2.5	1002	6	35	CS
	1800	7.9	81.3	2.5	1002	6	35	CS
	2100	8.1	81.2	2.5	1002	6	35	CS
28.11.25	0000	8.2	81.1	2.5	1002	6	35	CS
	0300	8.3	81.0	2.5	1002	6	35	CS
	0600	8.4	81.0	2.5	1002	6	35	CS
	0900	8.5	81.0	2.5	1002	6	35	CS
	1200	8.7	80.9	2.5	1002	6	35	CS
	1500	8.8	80.8	2.5	1002	6	35	CS
	1800	9.0	80.8	2.5	1002	6	35	CS
	2100	9.2	80.8	2.5	1001	7	35	CS

29.11.25	0000	9.4	80.7	2.5	1001	7	35	CS
	0300	9.6	80.7	2.5	1000	8	40	CS
	0600	9.9	80.7	2.5	1000	8	40	CS
	0900	10.1	80.6	2.5	1000	8	40	CS
	1200	10.4	80.6	2.5	1000	8	40	CS
	1500	10.5	80.6	2.5	1000	8	40	CS
	1800	10.7	80.6	2.5	1000	8	40	CS
	2100	10.8	80.6	2.5	1000	8	40	CS
30.11.25	0000	11.1	80.6	2.5	1000	8	40	CS
	0300	11.4	80.6	2.5	1000	8	40	CS
	0600	11.5	80.6	2.5	1001	7	35	CS
	0900	11.7	80.6	2.5	1002	6	35	CS
	1200	11.8	80.6	2.0	1003	5	30	DD
	1800	12.3	80.6	2.0	1003	5	30	DD
01.12.2025	0000	12.8	80.6	2.0	1003	5	30	DD
	0300	12.8	80.6	2.0	1003	5	30	DD
	0600	13.0	80.6	2.0	1003	5	30	DD
	1200	13.0	80.6	2.0	1003	5	30	DD
	1800	13.0	80.6	2.0	1003	5	30	DD
02.12.2025	0000	12.9	80.5	1.5	1004	4	25	D
	0300	12.8	80.5	1.5	1004	4	25	D
	0600	12.5	80.4	1.5	1004	4	25	D
	1200	12.3	80.3	1.5	1004	4	25	D
	1800	12.1	80.2	1.5	1005	3	20	D
03.12.2025	0000	Weakened into a Well Marked Low Pressure Area over North Tamil Nadu-Puducherry coasts and neighbourhood						

Kt: Knots (nautical mile per hour), 1 kt=1.85 kmph, D: Depression, DD: Deep Depression, CS: Cyclonic Storm

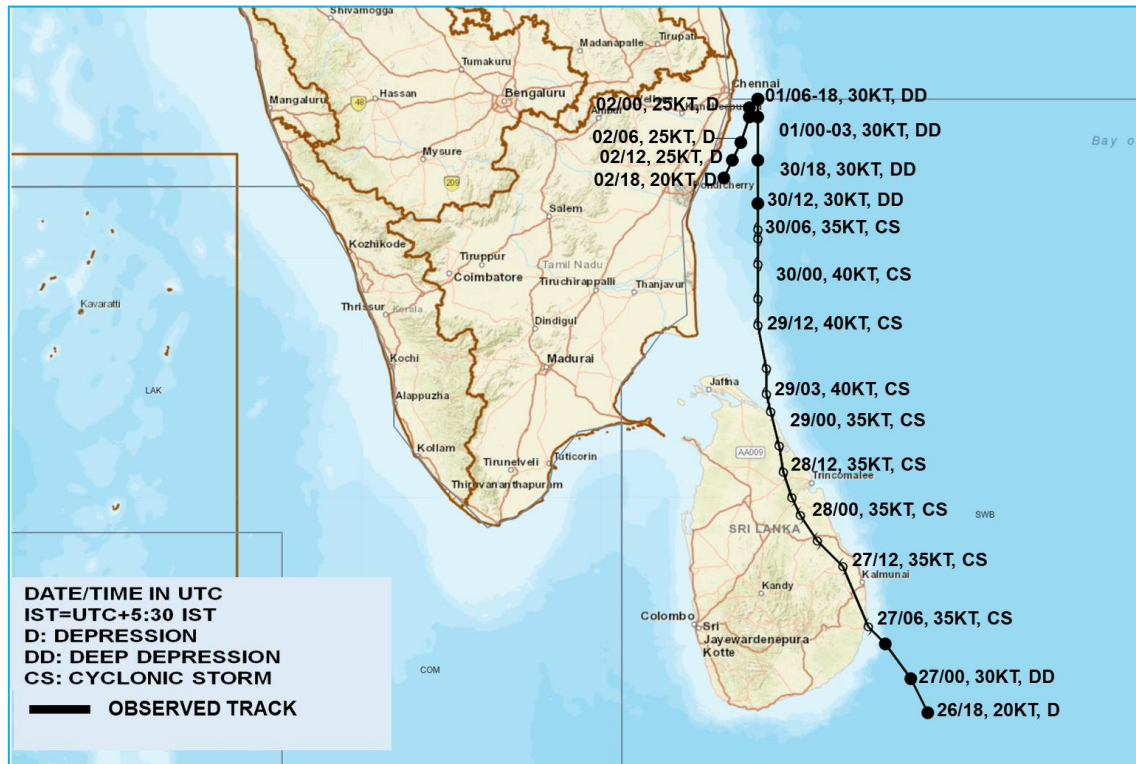


Fig. 1: Observed track of cyclonic storm “DITWAH” over Bay of Bengal during 26 November-03 December, 2025

2. Salient features:

- It is the first cyclone after 1967 (4-8 Dec.) which moved from South to North across Sri Lanka. There have been 21 cyclonic disturbances (depressions (maximum sustained wind speed (MSW) ≥ 17 kt and cyclones MSW ≥ 34 kt during 1960-2024 that crossed Sri Lanka coast (**Fig. 2a**). Out of these, there was a cyclone in 1967 (4-8 December) which moved from South to North across Sri Lanka. However, the cyclone in 1967 moved from central Sri Lanka to Southwest BoB near Tamil Nadu coast while moving northwards (**Fig. 2b**).

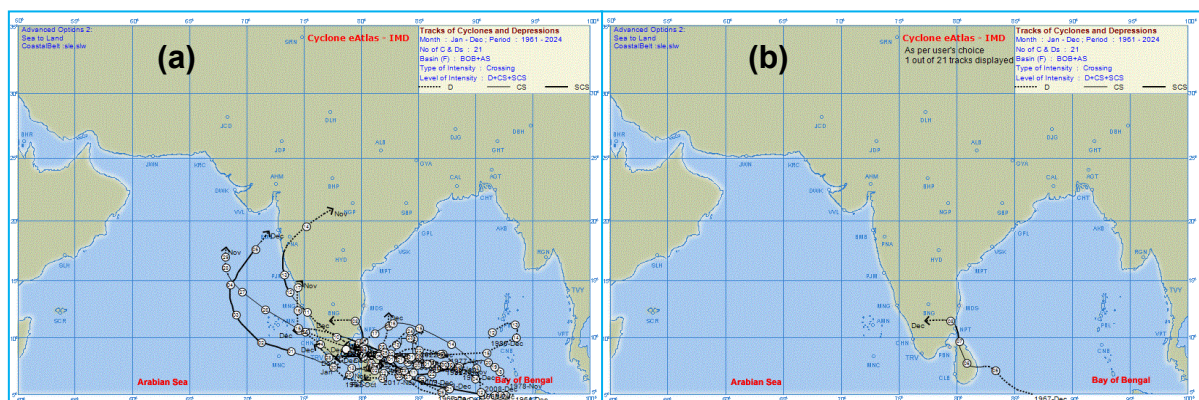


Fig.2: Tracks of (a) all cyclonic disturbances (MSW ≥ 17 kt) crossing Sri Lanka coast during 1961-2024 and (b) track of 1967 cyclone that moved south to north across Sri Lanka

- b) **Frequency of cyclones crossing Sri Lanka:** About 12 cyclones crossed Sri Lanka coast during 1961-2024 including 7 severe cyclonic storms (**Fig. 3**). All of these 12 cyclones crossed Sri Lanka coast in post monsoon season including six, five and one in the months of December, November, October respectively (**Fig.4**).

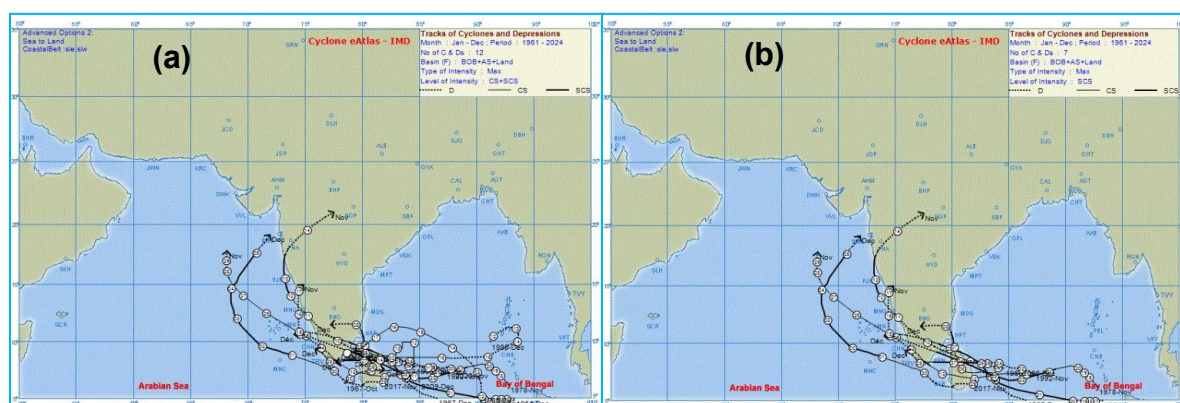


Fig.3: Tracks of (a) all cyclones (MSW≥34 knots) and (b) all severe cyclones (MSW≥48 knots) crossing Sri Lanka coast during 1961-2024

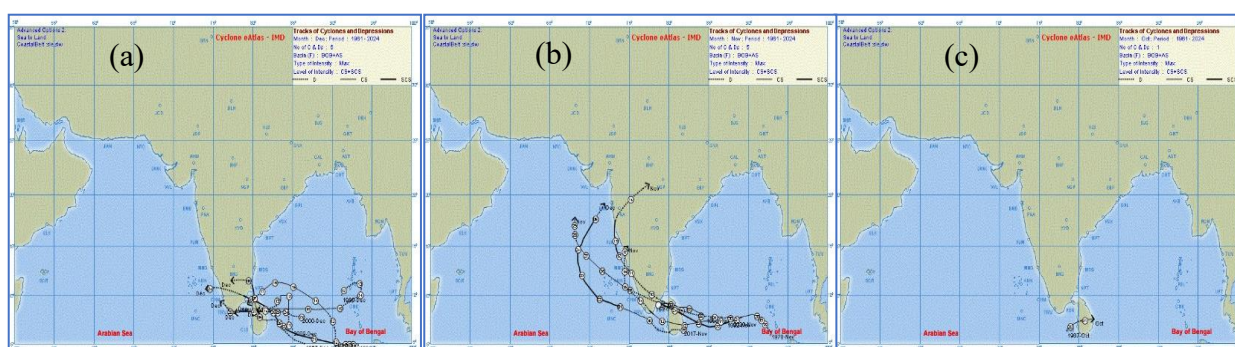


Fig.4: Tracks of cyclones crossing Sri Lanka coast in the months of (a) December, (b) November and (c) October during 1961-2024

- c) **Ditwah was the first cyclone** after cyclone Burevi (30 Nov-02 Dec) which crossed east Sri Lanka coast on 2nd Dec., 2020.
- d) **Simultaneous occurrence:** 2 cyclonic disturbances prevailed over the BoB during last week of November. Earlier such occurrence over BoB was seen in 2013, with two back to back cyclones Helen (19th-23rd November) and Lehar (23rd-28th November). Both crossed Andhra Pradesh coast. Helen developed over westcentral BoB on 19th Nov. early morning (0530 hrs IST/ 0000 UTC), intensified into cyclonic storm on 20th morning (0830 hrs IST/0300 UTC), into severe cyclonic storm on 21st Nov. early morning (0530 hrs IST/ 0000 UTC), crossed Andhra Pradesh coast on 22nd Nov. afternoon (1330-1430 hrs IST/ 0800-0900 UTC) and weakened into a well marked low pressure area over coastal Andhra Pradesh on 23rd early morning (0530 hrs IST/0000 UTC).

Another cyclone Lehar developed over South Andaman Sea in the evening (1730 hrs IST/ 1200 UTC), intensified into a cyclonic storm on 24th early morning (0530 hrs IST/ 0000 UTC), further into very severe cyclonic storm in the early hours (0230 hrs IST) of 26th Nov. (2100 UTC of 25th Nov.), weakened over westcentral BoB and crossed Andhra Pradesh coast as deep depression on 28th Nov. afternoon (around 1400 hrs IST/0830 UTC) on 28th Nov and weakened into a well marked low pressure area over coastal Andhra Pradesh and adjoining Telangana (**Fig. 5**).

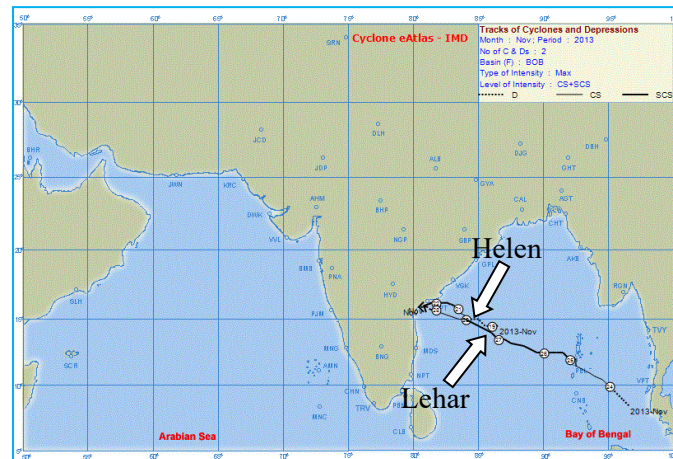


Fig. 5: Tracks of back to back cyclones Helen and Lehar in Nov., 2013

- e) **Unique Track:** The track of cyclone Ditwah was unique, as there has been no such track during the satellite era in the recorded history of IMD (since 1961 onwards) over that region. It had almost northward track from South of Sri Lanka to Southwest BoB close to North Tamil Nadu coast.
- f) **Recurving track:** Ditwah moved initially north-northwestwards till afternoon (1430 hrs IST/ 0900 UTC) of 29th Nov. and then northwards till morning (0830 hrs IST/ 0300 UTC) of 1st Dec., practically stationary till midnight (2330 hrs IST/ 1800 UTC) of 1st Dec., and then slowly recurved south-southwestwards till early morning (0530 hrs IST/ 0000 UTC) of 3rd Dec. (**Fig. 1 and Fig. 6a**). The system moved initially north-northwestwards and then northwards along the periphery of the ridge over the BoB around 12°N. However, as it remained almost stationary close to coast, persistent rainfall led to lowering of sea surface temperature. Further while moving northwards, it exhibited weakening due to land interactions, increase in cold dry air incursion from northwest and increased vertical wind shear. Consequently, the depth of convection reduced and the steering mechanism also changed. Thereafter, the north-northeasterly winds in the mid-tropospheric levels steered the system south-southwestwards.
- g) **Translational Speed:** Ditwah exhibited six hourly average translational speed of 9.8 kmph against normal of 15.5 kmph for cyclonic storms over the BoB in the post monsoon season during 1990-2013 (**Fig.6a**). Thus, it was a very slow moving cyclone. The average translational speed as it moved across Sri

Lanka during 27th noon (1130 hrs IST/0600 UTC) to 29th early morning (0530 hrs IST/0000 UTC) was even further slower (about 7 kmph), leading to extremely heavy rainfall activity over Sri Lanka during 25th to 29th November. Further, the interaction with orography over central Sri Lanka led to enhancement of rainfall activity.

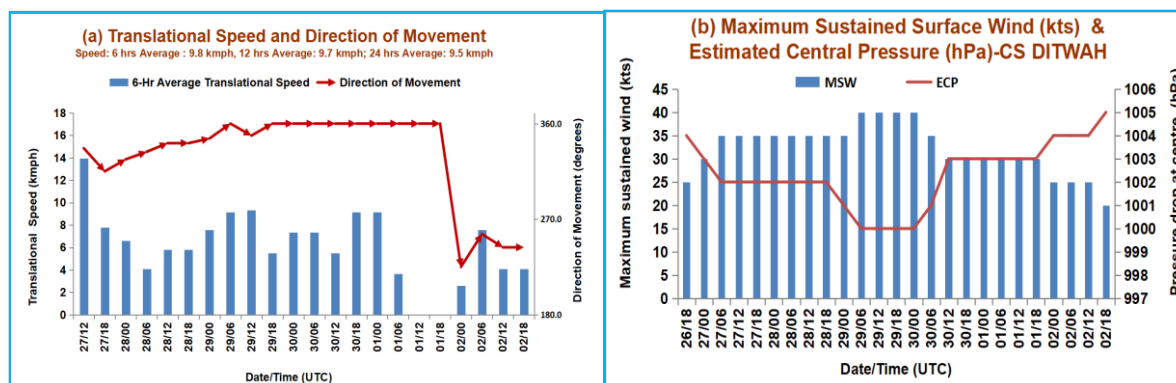


Fig. 6: Six hourly (a) average translational speed & direction of movement and (b) maximum sustained surface wind speed & estimated central pressure during life period of cyclone Ditwah

- h) **Life period:** The life period of the storm (depression to depression) was about 6 days and 06 hours against the long period average (LPA) (1990-2013) of about 3 days and 14 hours for CS category over the BoB during post-monsoon season. Thus, Ditwah had a life period longer than the average life period of cyclonic storms over the BoB during post-monsoon season.
- i) **Maximum sustained wind speed and estimated central pressure:** The system did not exhibit any rapid intensification/weakening during it's life cycle. The system reached it's peak intensity of 40 knots at 1130 hrs IST/0600 UTC of 29th November and maintained it till 0000 UTC of 30th November. The estimated central pressure during this period was 1000 hPa. The six hourly maximum sustained wind speed and estimated central pressure during the life cycle of Ditwah is presented in **Fig. 6b**.
- j) **Damage Potential and Power Dissipation Index:** The Accumulated Cyclone Energy (a measure of damage potential) and Power Dissipation Index (a measure of loss) in association with Ditwah were 1.74×10^4 knots² and 0.64×10^6 knots³ respectively against the normal of 1.0×10^4 knots² and 0.40×10^6 knots³ for CS during post monsoon season over the BoB based on the data during 1990-2020. As it was a slow moving cyclone with a longer life period, damage potential and losses due to Ditwah were higher than the long period average.
- k) **The storm surge** mostly impacted north Sri Lanka coast.

3. Monitoring of CS, “DITWAH”

India Meteorological Department (IMD) maintained round the clock watch over the north Indian Ocean and the cyclone was monitored since 13th November, about 13 days prior to formation of depression on 26th November. First information about likely genesis over the south BoB was issued in the extended range outlook issued on 13th November. The cyclone was monitored with the help of available satellite observations from INSAT 3DR, SCAT SAT (OCEAN SAT 3), ASCAT, microwave imageries, doppler weather Radar at Chennai & Karaikal, available ships, buoys & coastal observations in the region. Typical satellite and radar-based products utilised for monitoring the system are presented in **Fig. 7**.

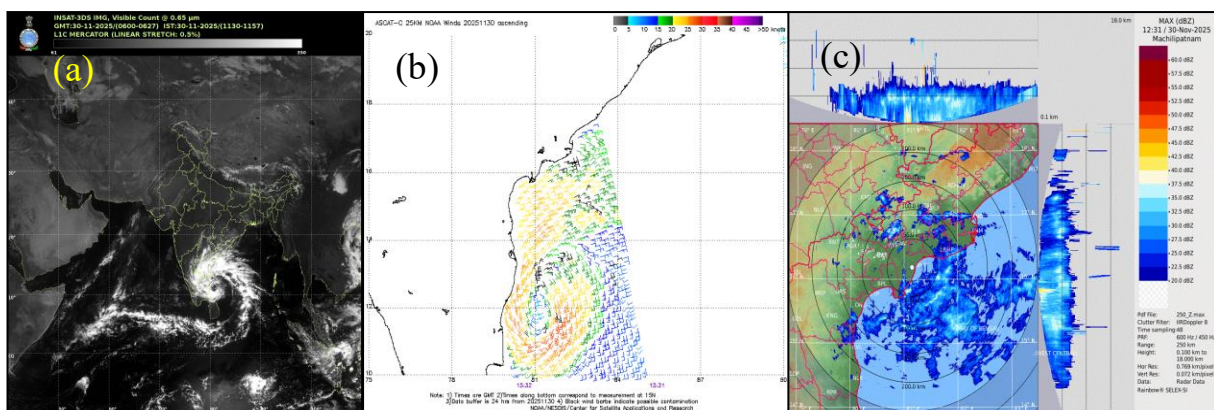


Fig. 7: (a) INSAT 3DR image of 1130 hrs IST (0600 UTC) of 30th November of (b) ASCAT image at 2100 hrs IST (1530 UTC) of 29th Nov, 2025 of cyclonic storm “DITWAH” over Bay of Bengal and (c) Machilipatnam Radar image at 1800 hrs IST (1230 UTC) of 30th November of cyclonic storm “DITWAH” over Bay of Bengal

Various global models and dynamical-statistical models run by Ministry of Earth Sciences (MoES) institutions including IMD, NCMRWF, IITM & INCOIS and guidance from models from various international agencies under bilateral arrangement were utilized to predict the genesis, track and intensity of the cyclone as well as associated severe weather. The forecasts were mainly based on multi-model ensemble technique developed by IMD which were further modulated by analysis of observations and forecasters intervention at IMD through their knowledge, experience and expertise. An indigenously developed digitized forecasting system of IMD (Decision Support System) was utilized for analysis and comparison of various observations and numerical weather prediction models guidance, decision making process and warning products generation.

4. Forecast performance:

- First information about likely cyclogenesis (formation of Depression) over the BoB was issued on 13th November on 26th November about 13 days ahead of formation of depression. Subsequently, the extended range outlook issued on 20th November (6 days ahead) indicated high (67-100%) probability of cyclogenesis (formation of Depression) over south BoB.
- In the tropical weather outlook issued at 1430 hrs. IST/ 0900 UTC of 23rd Nov., it was indicated that a low-pressure area would form over Comorin and adjoining southeast Sri Lanka around 25th Nov. with moderate probability of formation of depression by 25th morning (0830 hrs. IST/ 0300 UTC) and high probability of formation of depression by 26th morning (0830 hrs. IST/ 0300 UTC). Actually, the low-pressure area formed over Comorin & adjoining southeast BoB and Sri Lanka in the early morning (0530 hrs. IST/ 0000 UTC) of 25th Nov.
- The same was reiterated in the tropical weather outlook issued at 1430 hrs. IST/ 0900 UTC of 24th Nov.
- The tropical weather outlook issued at 1330 hrs. IST/ 0800 UTC of 25th Nov. on formation of low-pressure area over southwest Bay of Bengal and adjoining areas of south Sri Lanka and adjoining Equatorial Indian Ocean indicated north-northwestwards movement and likely intensification onto a well-marked low-pressure area by morning of 26th (0830 hrs. IST/ 0300 UTC) and depression by morning of 27th November (0830 hrs. IST/ 0300 UTC).
- Subsequently, the advisory issued in the morning of 26th Nov. (0930 hrs. IST/ 0400 UTC) on formation of well-marked low-pressure area over southwest BoB and adjoining areas of southeast Sri Lanka and Equatorial Indian Ocean predicted that the system would move north-northwestwards and intensify into a depression by morning of 27th Nov. (0830 hrs. IST/ 0300 UTC).
- The update issued in the noon of 26th (1230 hrs. IST/0700 UTC) indicated the well-marked low-pressure area to intensify into a depression by 27th morning and intensify further thereafter. It was predicted to move north-northwestwards across southwest BoB towards north Tamil Nadu and Puducherry coasts till 29th morning. Actually, it reached southwest BoB and adjoining north Sri Lanka and Tamil Nadu-Puducherry coasts as a cyclonic storm by 29th morning (0830 hrs. IST/ 0300 UTC).
- Actually, low pressure area formed over Comorin on 25th Nov. early morning (0530 hrs. IST/ 0000 UTC), well-marked low-pressure area over southwest BoB and adjoining southeast Sri Lanka on 26th early morning (0530 hrs. IST/ 0000 UTC) and depression over southwest BoB and adjoining Sri Lanka on 26th midnight (2330 hrs. IST/ 1800 UTC), Cyclonic Storm "DITWAH" over southwest BoB and adjoining Sri Lanka in the noon of 27th Nov. (1130 hrs. IST/ 0600 UTC).

- The first track and intensity forecast issued in the morning (0930 hrs IST/ 0400 UTC) of 27th Nov., indicated the system to reach peak intensification of 80-90 gusting to 100 kmph over southwest Bay of Bengal and along & off North Tamil Nadu-Puducherry coasts and adjoining North Sri Lanka from early morning of 29th Nov. to early morning of 30th Nov. Actually, Gale wind speed reaching 60 to 70 kmph gusting to 80 kmph was observed over south Sri Lanka on 27th Nov. and 65 to 75 gusting to 85 kmph was observed over central & north Sri Lanka during 27th to 28th Nov. Gale wind speed reaching 70-80 gusting to 90 kmph prevailed over southwest Bay of Bengal and south Tamil Nadu and adjoining north Tamil Nadu-Puducherry & north Sri Lanka during 29th morning to 30th morning.
- The first track and intensity forecast issued at 1540 hrs. IST of 27th indicated the system to gradually weaken into a deep depression while reaching near Tamil Nadu-Puducherry and adjoining south Andhra Pradesh coast by 01st Dec.
- The operational track forecast errors for 24, 48, 72 and 96 hours lead period were 49, 62, 60 and 100 km against long period average (2020-24) of 72, 111, 154 and 183 km respectively (**Fig. 8a**). The operational track forecast skill compared to climatological & persistence (CLIPER) based forecast for 24, 48, 72 and 96 hours lead period were 79, 88, 92 and 91% against long period average (2020-24) of 65, 74, 77 and 78 % respectively (**Fig. 8b**). **For all lead periods, the track forecast errors were markedly less than the long period average errors.**

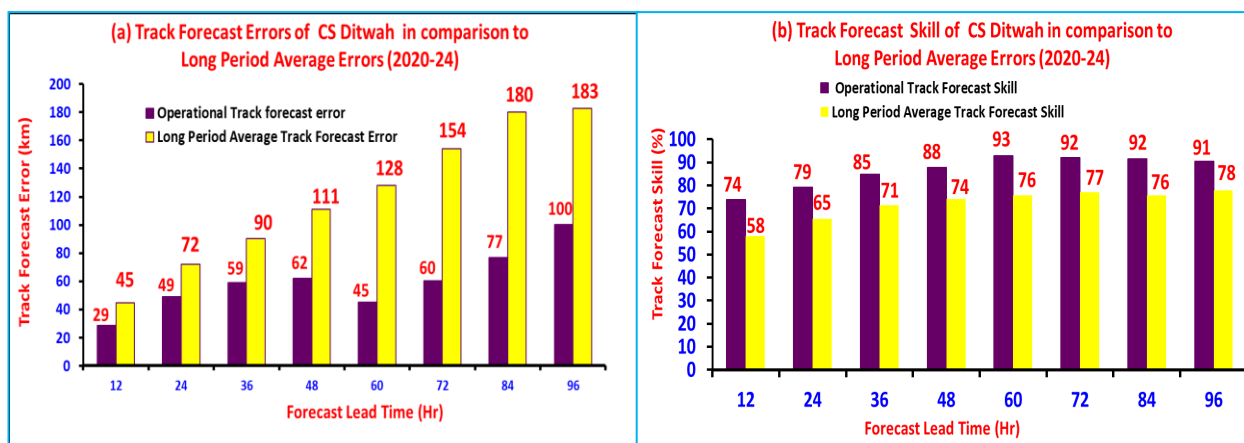


Fig.8: Operational track forecast (a) errors and (b) skills during CS Ditwah compared to long period average errors and skills during 2020-24

- The operational intensity forecast absolute errors (AE) for 24, 48, 72 and 96 hours lead period were 3.1, 3.0, 5.0 and zero kt against long period average (2020-24) of 5.9, 8.3, 9.8 and 15.7 kt respectively (**Fig. 9a**). The operational intensity forecast root mean square errors (RMSE) for 24, 48, 72 and 96

hours lead period were 4.4, 4.5, 5.8 and zero kt against long period average (2020-24) of 7.9, 13.3, 12.6 and 17.7 kt respectively (**Fig. 9b**). **For all lead periods, the intensity forecast errors were markedly less than the long period average errors.**

- The operational intensity forecast skill based on AE for 24, 48, 72 and 96 hours lead period was 76, 84, 86 and 100% against long period average (2020-24) of 57, 69, 79 and 81% respectively (**Fig. 10a**). The operational intensity forecast skill based on RMSE for 24, 48, 72 and 96 hours lead period was 81, 84, 86 and 100% against long period average (2020-24) of 57, 69, 79 and 81% respectively (**Fig. 10b**). **For all lead periods, the intensity forecast skills were markedly above the long period average skill.**

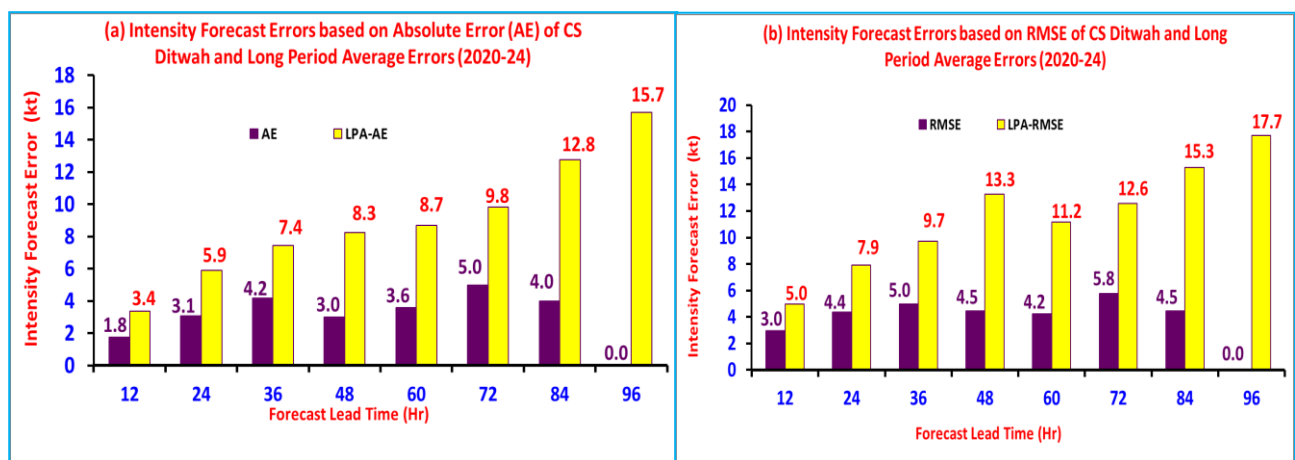


Fig.9: Operational intensity (wind) forecast (a) absolute errors and (b) root mean square errors (kt) during CS Ditwah compared to long period average errors during 2020-24

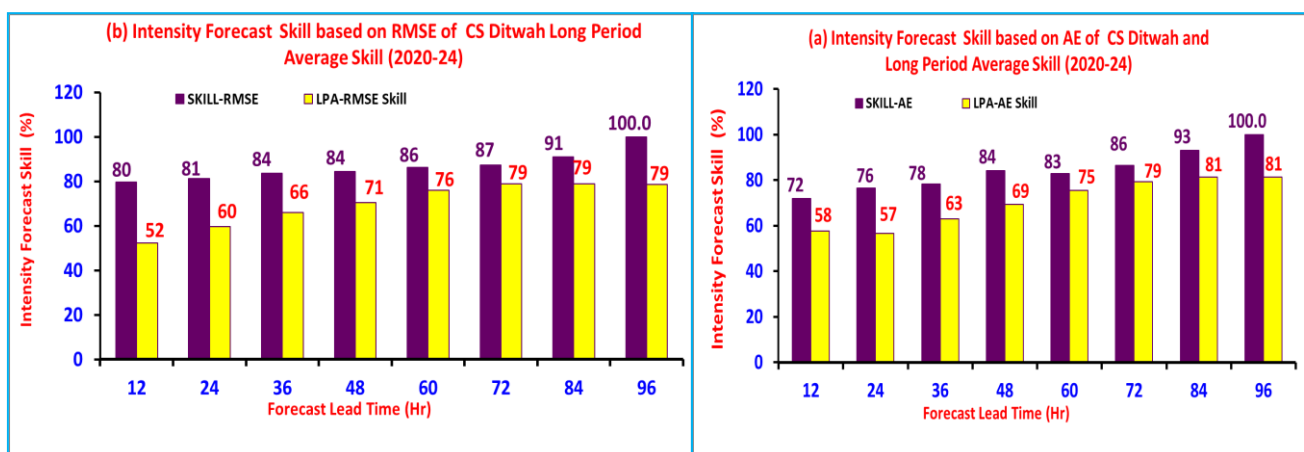


Fig.10: Operational intensity (wind) forecast skills based on (a) absolute errors and (b) root mean square errors (kt) compared to persistence-based errors during CS Ditwah vis-a-vis long period average skills during 2020-24

- The observed and forecast track (based on 0600 UTC/1130 hrs IST) demonstrating accuracy in track and intensity prediction is presented in **Fig.11**.

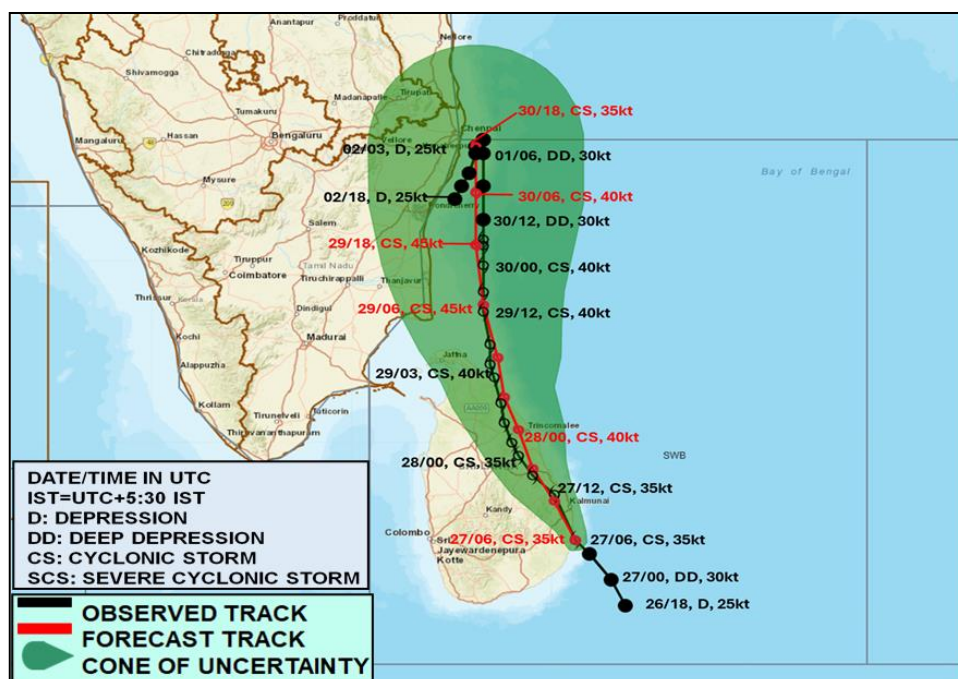


Fig. 11: Observed track and forecast track based on 1130 hrs IST (0600 UTC) of 27th Nov. of cyclonic storm “DITWAH” over Bay of Bengal indicating accuracy in track and intensity prediction

5. Realised Weather:

5.1. Realised rainfall:

The system caused extremely heavy rains and landslides in Sri Lanka during 26th to 28th Nov. Over Tamil Nadu, heavy to very rainfall at some places and extremely heavy rainfall (≥ 20 cm) at isolated places was observed on 28th, 29th November & 1st December and heavy to very heavy rainfall at isolated places on 2nd December. Heavy to very heavy rainfall at some places with isolated extremely heavy rainfall occurred over Coastal Andhra Pradesh & Rayalaseema on 2nd December. Satellite-gauge merged rainfall plots showing realised winds during cyclone Ditwah is presented in **Fig. 12**.

24 hours cumulative realised rainfall (≥ 7 cm) ending at 0830 hrs IST (0300 UTC) of date as recorded by various observatories in India is presented below:

29.11.2025:

Tamil Nadu, Puducherry & Karaikal: Kodiayakarai 25; Vedaranyam) 18; Velankanni 13; Nagapattinam, Tirupoondi 12 Each; Thalaignayer 11; Karaikal, Pamban 10 Each; Rameswaram Illayangudi Mandapam Kayalpattinam, Thirukuvalai, Thangachimadam 7 each.

30.11.2025:

Tamil Nadu, Puducherry & Karaikal: Karaikal 19; Sembanarkoil 17; Mayiladuthurai, Nagapattinam 15 each; Mayiladuthurai, Sirkali, Tiruvarur 14 Each; Tondi, Tirupoondi, Thangachimadam 13 Each; Sethiathope, Vedaranyam, Nannilam, Velankanni, Kodiayakarai, Kurungulam 12 each; Srimushnam, Thiruthuraipoondi, Tiruvarur AWS, Thalaigayyer, Thirukuvalai, Neivasal Thenpathi, Echanviduthi 11 each; Cuddalore, Parangipettai, Kollidam, Tiruvadanai, Orthanad, Mannargudi, Mimisal, Collectorate, Theerthandathanam 10 each; Chidambaram, Rameswaram, Pattukottai, Peravurani, Thiruvaidaimaruthur, Needamangalam, Valangaiman, Manalmedu, Avudayarkoil, Vattanam, Ayyampettai, Budalur, Vadakuthu, Ayinkudi, Nagudi, Vettikadu 9 Each; Arantangi, Pamban, Madukkur, Thanjavur, Tiruvaiyaru, Sivakasi, Aduthurai AWS, Thanjavur, Bhuvanagiri, Mandapam, Manjalaru, Annamalai Nagar, Lalpet, Nandhiyar Head 8 each; K.M.Koil, Mahabalipuram, Tarangambadi, Puducherry, Alangudi, Pudukottai, Devakottai, Adirampatnam, Vallam, Muthupet, Pandavaiyar Head, Kodavasal, Kothavacherry, Lower Anaicut 7 each.

01.12.2025:

Coastal Andhra Pradesh & Yanam: Kavali 11

02.12.2025:

Tamil Nadu, Puducherry & Karaikal:

Ennore Zone1 26, Parrys 25, Ice House 22, Ponneri 21; Manali New Town 21, Ennore, Perambur, Chennai Collectorate BU 20 each; DGP Office, Chennai(n), Red Hills 18 each; Gummidipoondi, Ayanavaram Taluk Office, CD Hospital Tondaipe 17 each; Puzhal 16; Cholavaram, MGR Nagar 14 Each; Anna University 13; Anna Udy, Sholinganallur 11 each; Chennai AP, Sivakasi, Hindusthan University, Alandur, Avadi 9 each; Taramani 8; Kolapakkam, Ambathur 7 each.

03.12.2025:

Rayalaseema: Chittamur 27, Kota 23, Gudur, Satyavedu, Tada 9 each; Sullurpeta 7;

Tamil Nadu, Puducherry & Karaikal: Hindusthan University 15; Chetpet 13; Tirumayam, Thamarapakkam 12 each; Tiruvarur, Satyabama Udy, Thirukalukundram, Kudimiyamalai 11 each; Kelambakkam, Mahabalipuram, Illuppur, Red Hills, Ulundurpet, Vadakuthu 10 Each; Pollachi, Tondi, Cholavaram, Ponneri, Tiruvarur, Gummidipoondi, Ambathur, Perambur, Neivasal Thenpathi 9

Each; Needamangalam, Uthukottai, RSCL Vallam 8 each; Virudachalam, Karaikal, Mannargudi, Sholinganallur, DSCL Eraiyur, Avadi, Ranipet 7 each;

Coastal Andhra Pradesh & Yanam: Edagali 24, Nellore 9;

Kerala & Mahe: Vannamada 7.

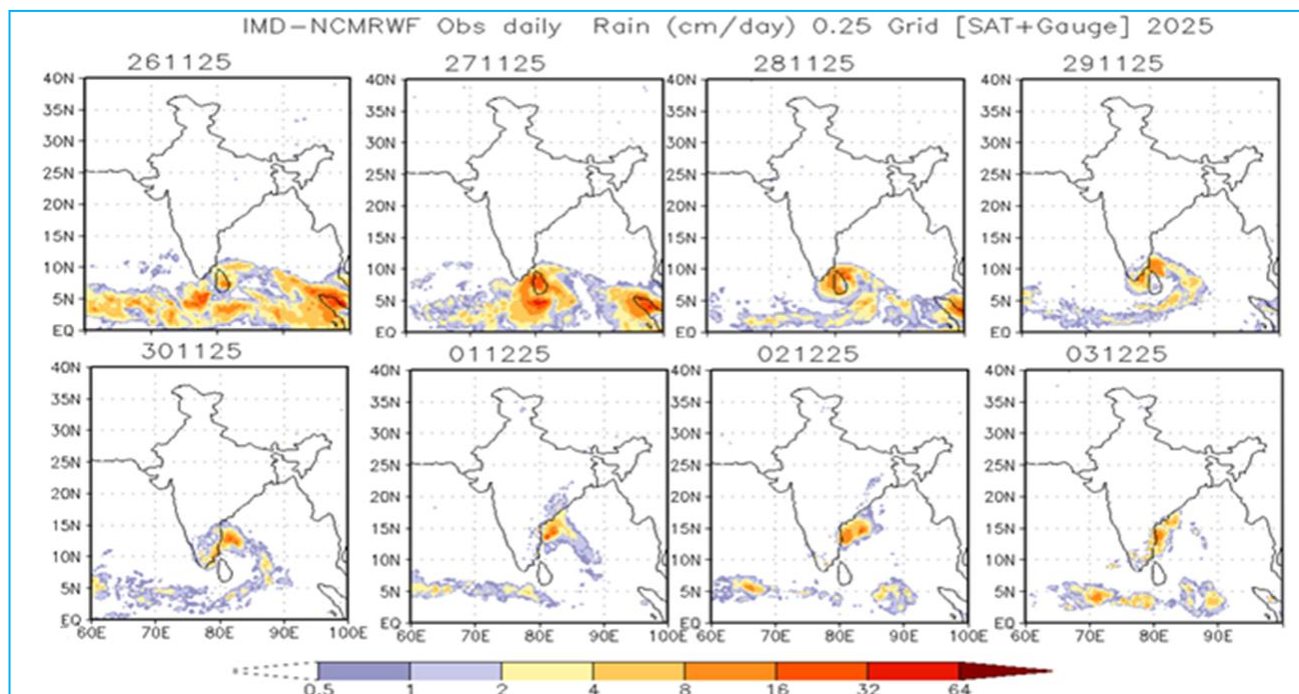


Fig. 12: IMD-NCMRWF satellite-gauge merged rainfall plots during 26th Nov. to 3 Dec., 2025.

5.2. Realised wind:

Gale wind speed reaching 60 to 70 kmph gusting to 80 kmph was observed over south Sri Lanka on 27th Nov. and 65 to 75 gusting to 85 kmph was observed over central & north Sri Lanka during 27th to 28th Nov. Gale wind speed reaching 70-80 gusting to 90 kmph prevailed over southwest Bay of Bengal and south Tamil Nadu and adjoining north Tamil Nadu-Puducherry & north Sri Lanka during 29th morning to 30th morning. The realised wind during CS Ditwah is presented in **Fig. 13**.

5.3. Realised storm surge:

Storm surge of height about 1.0 to 1.5 m above the astronomical tide inundated the low-lying coastal areas of north Sri Lanka during 27th morning till 29th evening. Storm surge of height 0.5 m – 1.0 m above astronomical tide inundated low lying areas of coastal Tamil Nadu – Puducherry on 29th November. Significant wave height of 1.8 m was observed near Pamban bridge on 29th Nov.

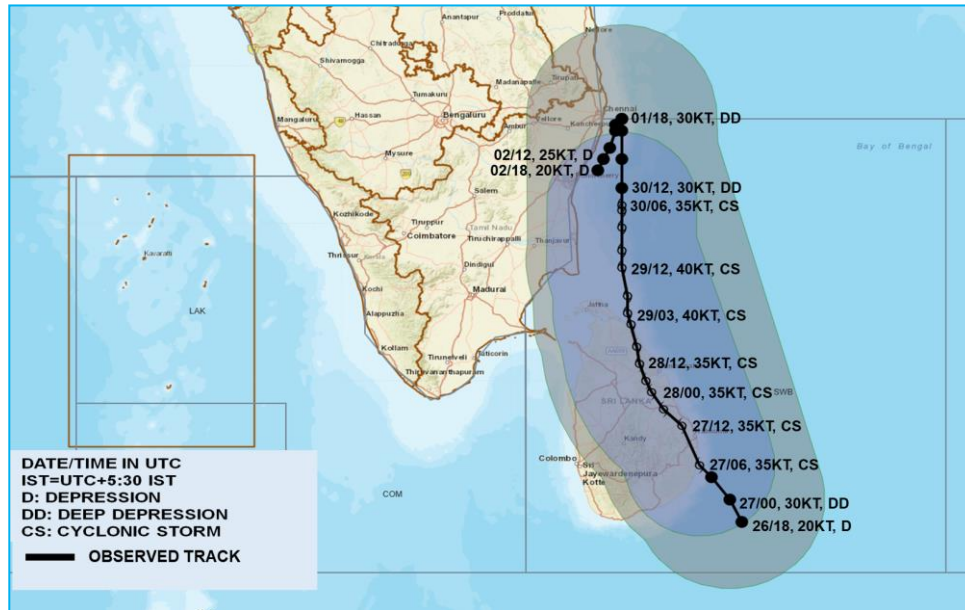


Fig. 13: Realised maximum sustained wind during CS Ditwah

6. Bulletins issued by Cyclone Warning Division, New Delhi

- **Track, intensity and landfall forecast:** IMD continuously monitored, predicted and issued bulletins containing track, intensity, and landfall forecast for +06, +12, +18, +24, +36 and +48... +96 hrs lead period commencing from 27th November morning (0530 hours IST/0000 UTC) till the system weakened into a depression. The above forecasts were issued from the stage of depression onwards along with the cone of uncertainty in the track forecast five times a day and every three hours during the cyclone period.
- **Cyclone structure forecast for shipping and coastal hazard management:** The radius of maximum wind and radii of MSW ≥ 28 , ≥ 34 and ≥ 50 knots wind in four quadrants of cyclone was issued every six hourly, commencing from 27th November morning (0530 hours IST/0000 UTC) giving forecast for +06, +12, +18, +24, +36 and +96 hrs lead period.
- **Four stage Warnings:**
 - ❖ **Pre-cyclone Watch (Yellow message)** for Tamil Nadu-Puducherry and adjoining Andhra Pradesh coasts issued at 0945 hrs. of 27th November indicated the system to reach near Tamil Nadu-Puducherry and adjoining South Andhra Pradesh coasts with maximum sustained wind speed of 80-90 kmph gusting to 100 kmph by 30th morning.
 - ❖ **Cyclone Alert (Orange message)** for Tamil Nadu-Puducherry and adjoining south Andhra Pradesh coasts issued on 27th Nov (2245 hrs. IST) indicated the system to reach near Tamil Nadu-Puducherry and adjoining south Andhra Pradesh coasts with maximum sustained wind speed of 80-90 kmph gusting to 100 kmph by 30th morning.

- ❖ **Cyclone Warning (Red message)** for Tamil Nadu-Puducherry and adjoining south Andhra Pradesh coasts was issued on 1300 hrs. IST of 29th Nov.
- ❖ **Regular messages** through CAP, SMS, whatsapp, facebook, twitter, IMD websites were issued since 23rd Nov.
- **Adverse weather warning bulletins:** The tropical cyclone forecasts alongwith expected adverse weather like heavy rain, gale wind and state of sea was issued with every six/three hourly update to central, state and district level disaster management agencies including Ministry of Home Affairs (MHA), National disaster Response Force (NDRF), National Disaster management Agency (NDMA) for all concerned states along the east coast of India including Tamil Nadu & Puducherry, Andhra Pradesh, Kerala, West Bengal and Odisha. The bulletins also contained the suggested actions for disaster managers and general public in particular for fishermen, ports and off & along shore activities. These bulletins were also issued to Defence including Indian Navy & Indian Air Force, NDRF, Indian Coast Guard, ports, Shipping, fishery, Railways, surface transport and aviation authorities.
- **Warning graphics:** The graphical display of the observed and forecast track with cone of uncertainty and the wind forecast for different quadrants were disseminated by email and uploaded in the RSMC, New Delhi website (<http://rsmcnewdelhi.imd.gov.in/>) regularly. The adverse weather warnings related to heavy rain, gale wind & fishermen warnings were also presented in graphics alongwith colour codes in the website.
- **Warning and advisory through social media:** Daily updates (every three hourly or whenever there was any significant change in intensity/track/landfall) were uploaded on Facebook and Twitter during the life period of the system since the formation of low-pressure area over Comorin Area.
- **Press Conference, Press Release and Media briefing:** Press and electronic media were given daily updates since inception of system through press release, e-mail, website, video capsule by DGM and SMS.
- **Warning and advisory for marine community:** The three/six hourly Global Maritime Distress Safety System (GMDSS) bulletins were issued by the Marine Weather Services Division at New Delhi and bulletins for maritime interest were issued by Area cyclone warning centres of IMD at Kolkata and Area Cyclone Warning Centre at Chennai, Cyclone warning centres at Visakhapatnam and Bhubaneswar to ports, fishermen, coastal and high sea shipping community.
- **Fishermen Warning:** Regular warnings for fishermen for deep Sea of Bay of Bengal were issued since 22nd November.
- **Advisory for international Civil Aviation:** The Tropical Cyclone Advisory Centre (TCAC) bulletin for International Civil Aviation were issued every six hourly to all meteorological watch offices in Asia Pacific region for issue of significant meteorological information (SIGMET). It was also sent to Aviation Disaster Risk Reduction (ADRR) centre of WMO at Hong Kong.

- **Diagnostic and prognostic features of cyclone:** The prognostics and diagnostics of the systems were described in the RSMC bulletins.
- **Director General of Meteorology** gave a presentation on the status of CS Ditwah during the National Crisis Management Committee Meeting chaired by Hon'ble Cabinet Secretary.

Statistics of bulletins issued in association with cyclonic storm Ditwah is presented in **Table 2**.

Table 2: Bulletins issued by Cyclone Warning Division, New Delhi

S. No.	Bulletin type	No. of Bulletins	Issued to
1	National Bulletin	51	1. IMD's website, RSMC New Delhi website 2. FAX and e-mail to Control Room Ministry of Home Affairs & National Disaster Management Authority, Cabinet Secretariat, Minister of Science & Technology, Secretary MOES, Headquarter Integrated Defence Staff, Director General Doordarshan, All India Radio, PIB MOES, DG National Disaster Response Force, Director Punctuality, Indian Railways, Secretaries to Govt of India for Surface Transport, Mines, Agriculture, Ports, Shipping & Waterways, Fishery, Aviation, Power, Telecommunication, Petroleum & Natural Gas etc. and Chief Secretary to Government of Tamil Nadu, Puducherry, Andhra Pradesh, Odisha, West Bengal, Andaman & Nicobar Islands, Kerala, Lakshadweep.
2	Special Bulletin from DG IMD	6	High level disaster managers by email and FAX including Principal secretary to Prime Minister, Home Secretary, NDMA, NDRF, Secretaries of various Ministries & concerned states and disaster management agencies mentioned in SN1.
3	RSMC Bulletin	57	1. IMD's website 2. WMO/ESCAP member countries through GTS and E-mail
4	Flash Flood Guidance Bulletin	25	Email to National level disaster managers, Central Water Commission, Ministry of Home Affairs, Ministry of Water Resources, South Asian countries including Sri Lanka Meteorological Department, Flood Met Offices, social media, RSMC & Mausam website
5	GMDSS Bulletins	30	1. IMD website, RSMC New Delhi website 2. Transmitted through WMO Information System (WIS) to Joint WMO/IOC Technical Commission for Ocean and Marine Meteorology (JCOMM)
6	Tropical Cyclone Advisory Centre Bulletin	15	1. Met Watch offices in Asia Pacific regions and middle east through GTS to issue Significant Meteorological information for International Civil Aviation 2. WMO's Aviation Disaster Risk Reduction (ADRR), Hong Kong through ftp 3. RSMC website

7	Tropical Cyclone Vital Bulletin	15	Modelling group of IMD, National Centre for Medium Range Weather Forecasting Centre (NCMRWF), Indian National Centre for Ocean Information Services (INCOIS), Indian Institute of Technology (IIT) Delhi, IIT Bhubaneswar etc.
8	Warnings through SMS	Frequently	SMS to disaster managers at national level and concerned states (every time when there was change in track, intensity and landfall characteristics) (i) 3,42,476 to General Public by IMD Headquarters
9	Warnings through Social Media	51 times	Cyclone Warnings were uploaded on social networking sites (Facebook and Tweeter) since inception to weakening of system (every time when there was change in track, intensity and landfall characteristics).
10	Warnings through WhatsApp	51 times	Warnings and bulletins were shared through WhatsApp with Disaster managers, media, WMO/ESCAP Panel member countries
11	Press Release	6	Disaster Managers, Media persons by email and uploaded on website
12	Press Briefings	Daily	Regular briefing frequently

7. Acknowledgement:

IMD and Regional Specialised Meteorological Centre (RSMC) New Delhi duly acknowledge contribution from WMO and WMO/ESCAP member countries including Sri Lanka in dissemination of bulletins and warnings associated with CS Ditwah. IMD and RSMC New Delhi also acknowledge the contribution from all the stake holders and disaster management agencies at National and State level who contributed to the successful monitoring, prediction and early warning service of CS Ditwah. We acknowledge the contribution of all sister organisations of Ministry of Earth Sciences including National Centre for Medium Range Weather Forecasting Centre (NCMRWF), Indian National Centre for Ocean Information Services (INCOIS), National Institute of Ocean Technology (NIOT), Indian Institute of Tropical Meteorology (IITM) Pune. The support from various Divisions/Sections of IMD including Area Cyclone Warning Centre (ACWC) Kolkata & Chennai, Cyclone warning Centre Visakhapatnam, Meteorological Office Port Blair, Meteorological Centre Amravati, all coastal observatories along the east coast of India and Doppler Weather Divisions at Chennai and Karaikal. The contribution from Numerical Weather Prediction Division, Satellite and Radar Division, Surface & Upper air instruments Divisions, New Delhi, National Weather Forecasting Centre (NWFC) New Delhi and Information System and Services Division at IMD is also duly acknowledged.
